

section .data

array db 11h,59h,33h,22h,44h

msg1 db 10,"ALP to find the largest number in an array",10

msg1_len equ \$ - msg1

msg2 db 10,"The Array contains the elements : ",10

msg2_len equ \$ - msg2

msg3 db 10,10, "The Largest number in the array is : ",10

msg3_len equ \$ - msg3

section .bss

counter resb 1

result resb 4

%macro write 2

mov rax,1

mov rdi,1

mov rsi,%1

mov rdx,%2

syscall

%endmacro

section .text

global _start

_start:

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write msg1 , msg1_len
write msg2 , msg2_len

mov byte[counter],05
mov rsi,array
next:  mov al,[rsi]
      push rsi
      call disp
      pop rsi
      inc rsi
      dec byte[counter]
      jnz next

write msg3 , msg3_len

mov byte[counter],05
mov rsi, array
mov al, 0      ; al is an 8 bit register , al stores max
repeat: cmp al,[rsi] ;cmp opr1 , opr2 : opr1 - opr2
      jg skip
      mov al,[rsi]
skip:  inc rsi
      dec byte[counter]
      Jnz repeat

call disp

mov rax,60
mov rdi,1

```

syscall

disp:

mov bl,al ;store number in bl
mov rdi, result ;point rdi to result variable
mov cx,02 ;load count of rotation in cl

up1:

rol bl,04 ;rotate number left by four bits
mov al,bl ;move lower byte in dl
and al,0fh ; get only LSB
cmp al,09h ;compare with 39h
jg add_37 ;if grater than 39h skip add 37
add al,30h
jmp skip1 ;else add 30

add_37: add al,37h

skip1: mov [rdi],al ;store ascii code in result variable

inc rdi ;point to next byte
dec cx ;decrement the count of digits to display
jnz up1 ;if not zero jump to repeat

write result , 4

ret